

NeuroPulse AI: Electromyography (EMG) Profile

Yellow = cleaned signal | Green = muscle strength | Red = muscle active shadow

Patient: Patient Name | **ID:** NP-20260522-6079 | **Date:** 22-05-2026 21:45

Avg Strength 13.5	Active Time 1.4%	Major Spikes 3	Signal Quality Fair
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1. Clinical Summary

Session Score	13/100
Symmetry Score	5.7%
Side Deficit	94.3% (Left lower)
Lowest Activity Channel	Channel 3
Fatigue Drop	High (100.0% decline)
Signal Quality	Fair

2. 30-Second Graph Snapshot



3. Muscle-wise Quantitative Summary

Muscle / Channel	Avg	Peak	Active %	Quality
Channel 1	2.8	812.3	0.9%	Low
Channel 2	0.1	57.2	0.2%	Low
Channel 3	0.0	13.8	0.0%	Low
Channel 4	51.2	2781.6	4.7%	Low

4. Local Data-Based EMG Findings

Signal Reasoning (Quantitative Text Analysis):

- * Session Score: 31/100. Signal Quality: Fair.
- * Side Deficit: 99.1%. Symmetry Score: 0.9%. Marked left-side deficit detected.
- * Lowest Activity: Channel 2. Highest Peak: Channel 4.
- * Fatigue Drop: Low (0.0% decline from early to recent activity).
- * Channel Balance: left-side channel balance 0%; right-side channel balance 0%.
- * Suggested focus: compare the two right-side muscles/channels and train the weaker activation pattern.
- * Selected affected side: Select Affected Side. Non-diagnostic rehab monitoring support only.

Recorded Session Findings:

- * Recording summary: 4750 EMG samples captured over approximately 60 seconds.
- * Session Score: 13/100. Signal Quality: Fair.
- * Side Deficit: 94.3%. Symmetry Score: 5.7%. Impression: marked left-side activation deficit. Bilateral EMG symmetry is reduced and requires targeted rehab attention.
- * Lowest Activity: Channel 3 showed the lowest average activation (0.0).
- * Highest Peak: Channel 4 showed the highest peak activation (2781.6).
- * Channel Balance: left-side channel balance 2.6%; right-side channel balance 0.0%.
- * Fatigue Drop: High fatigue pattern with 100.0% decline from early to recent activity.
- * Sustained activation: Channel 1 0.9%, Channel 2 0.2%, Channel 3 0.0%, Channel 4 4.7% above threshold.
- * Clinical note: Affected side was not selected; interpretation is limited to channel activity and symmetry. This is a non-diagnostic rehab monitoring prototype that provides quantitative EMG feedback for physiotherapy support.

5. Clinical Observations & Notes

Summary Notes: N/A

Comparative Analysis:

- Left Side Avg: 1.5
- Right Side Avg: 25.6
- Symmetry Ratio (L/R): 5.7%
- Most Dynamic Channel: Channel 4